

Has the Structure of Risk Changed Since COVID-19?

A walk through volatility, diversification and the way shocks travel

Empirical Methods in Finance, Project 2

Professor Florian Ielpo

University of Lausanne | Master of Science in Finance

Group members:

Danny Al Fallouji

Selim Grar

Diona Avdija

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Abstract. We ask a simple question: after the COVID-19 shock of March 2020, did the structure of financial risk actually change, or did markets just go through a violent but temporary scare? We use 14 daily series from January 1990 to April 2026 and a small set of tools from the course, picked one by one as the data tell us what to look at next. Our answer is *partially*. The average level of volatility and the average level of correlation did not move. What moved is the *dynamics*: volatility shocks now hit a bit harder and fade faster (GARCH likelihood-ratio test, $p = 0.003$), and cross-asset correlations react faster to shared shocks (DCC test, $p < 0.001$). Factor concentration tightened only weakly: under honest inference the average number of independent risk factors did not fall significantly, but its variability did. Tail losses look heavier, yet our post-COVID sample is too short to confirm it at the 1% tail. The static link between volatility and correlation did not change. The practical message: update the parts of a risk model that describe *how fast* risk moves, not the parts that describe its average level.

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1. Introduction and Related Work

In a few weeks of March 2020 the S&P 500 fell by more than a third, oil futures briefly traded below zero, and assets that normally drift apart suddenly dropped together. It was the sharpest shock to markets since 2008. Five years on, a fair question for a risk manager is whether that episode left a mark on the *structure* of risk, or whether markets simply passed through an extreme moment and went back to behaving as before.

We need to say what “structure of risk” means before we can test it. We take it to be the shape of the joint distribution of asset returns, described by a few features a portfolio manager actually cares about: how big individual shocks are, how heavy the tails are, how much assets move together, how many independent risk factors are left for diversification, and how shocks travel and persist over time. This is deliberately a list of distinct features, because a market can keep the same *average* correlation while its tails get heavier or its factor structure gets tighter. Lumping everything into one number would hide exactly the kind of change we are looking for.

Why does it matter in practice? Risk limits, diversification budgets and regulatory capital are all calibrated on history. If the structure of risk shifted after 2020, a model fed with pre-2020 parameters will give biased answers, and the bias will be silent. So the question is not academic curiosity; it decides whether old calibrations can still be trusted.

The literature gives us both the hypotheses and the warnings. Longin and Solnik (2001) and Forbes and Rigobon (2002) show that correlations rise in turbulent periods, and that this is partly mechanical: when volatility is high, measured correlation goes up almost by construction. That tells us to separate a genuine COVID effect from this mechanical link. Engle (1982) and Bollerslev (1986) give us the ARCH/GARCH family for volatility clustering, and Glosten et al. (1993) the asymmetric version where bad news raises volatility more than good news. For correlation that changes through time, Engle (2002) provides the DCC model. Embrechts et al. (1997) give the extreme-value tools for the tail. For regimes we lean on Hamilton (1989), and Ang and Bekaert (2002) add the useful point that correlations are themselves regime-dependent. Bae et al. (2003) push the idea further: diversification collapses when assets crash together. The COVID papers came fast. Baker et al. (2020) show the uncertainty shock was historically large, but most of that early work stopped at the crisis weeks and never asked whether the change stuck. That longer-horizon question is the gap we try to fill.

Our plan is a single thread, not a pile of tests. The preliminary diagnostics choose our tools, and we then walk three blocks where each one answers the question the previous block leaves open. Block A is about the marginals: how big and how heavy are individual shocks? We answer it with realised volatility, GARCH and extreme-value theory. Block B asks whether diversification still works, using average correlation, factor concentration and a DCC model. Block C is about the dynamics, that is, how shocks travel and persist, and there we use a regression, a small VAR and a regime model. We give away the ending now. The change is partial, and it sits in the dynamics, not in the average levels.

2. Data

Everything comes from the provided `Data.xlsx` file: 14 daily series from January 1990 to April 2020, about 9,600 trading days. They cover five asset classes: five equity indices (S&P 500, EuroStoxx 50, Hang Seng, MSCI Emerging Markets, SMI), two ten-year sovereign yields (US and Germany), two commodities (WTI oil and gold), three exchange rates (EUR/USD, USD/JPY, USD/CHF) and two US credit indices (investment grade and high yield). We use no outside data.

For the price series we take log-returns, $r_t = \ln P_t - \ln P_{t-1}$. The two yields are rates, not prices, so a log-return on them has no meaning; we use first differences, $\Delta y_t = y_t - y_{t-1}$. Where a market is closed for a holiday while others trade, we forward-fill for at most two days to keep the calendars aligned. From the five equity indices we build one equal-weight *equity composite*, which is the “market” series we use throughout, so that the volatility, correlation and regression results all describe the same object.

One honest caveat about the data. The equity composite mixes indices that close at different hours (New York, Europe, Hong Kong). This non-synchronous trading pushes measured same-day correlations down a little, and the forward-fill adds a few zero-return days that do the same. Both effects are small at the 60-day horizon we work with, but we keep them in mind when reading the correlation results.

We split the sample at March 2020. The choice is easy to defend: the WHO declared the pandemic on 11 March, the Fed cut to zero on 15–17 March, equities had their worst week since 2008, and oil went negative in April. But “defensible date” is not the same as “clean break”. A Chow test on the baseline correlation–volatility regression gives $F = 0.78$ ($p = 0.46$): there is no sharp jump at that exact month. A CUSUM test on the same residuals rejects stability over the whole sample ($p < 0.001$). Read together, the picture is a gradual drift rather than one clean structural break, which is worth remembering when we interpret the COVID dummy later.

The split is also very unbalanced: 347 pre-COVID months against 73 post-COVID months. The pre-COVID window is not one regime; it averages the dot-com bust, the 2008 crisis and the euro crisis. So “pre-COVID” really means “the long and varied past”. This imbalance is the single biggest limitation of the study, and it shapes how carefully we read every comparison.

3. Methodology

This section explains the tools and what we expect from each before seeing the numbers. The logic is that the diagnostics decide which tools are appropriate, so we start there.

3.1. Diagnostics that choose the tools

Three quick tests set up everything else. The Jarque–Bera test asks whether returns look normal; we expect a clean no, with fat tails. A Ljung–Box test on *squared* returns asks whether volatility clusters; we expect yes. And the ADF and KPSS tests ask whether our monthly series are stationary, which decides how we are allowed to run regressions. Fat tails point us to a Student- t GARCH and to extreme-value theory rather than to normal-based measures. Clustering justifies GARCH. And if the correlation series turns out to be highly persistent, the course’s own warning about spurious regression (Phillips, 1986: persistent series regressed in levels give high R^2 and autocorrelated residuals that mean nothing) tells us to lean on a lagged-dependent (ARX) specification rather than a plain level regression.

3.2. An inference rule we use everywhere

Our monthly risk measures are built on overlapping 60-day windows, so neighbouring months share most of their data and are strongly autocorrelated. A standard t -test treats them as independent and would report tiny, dishonest p -values. To avoid that trap we use a *stationary block bootstrap* for every pre/post comparison: we resample contiguous blocks of months, which keeps the autocorrelation, and read the p -value from the resampled distribution. We also report each comparison against two benchmarks, the full pre-COVID history and the recent 2015–2020 calm. That split lets us separate “different from the long past” from “different from the recent quiet years”.

3.3. Block A – marginals

We measure the monthly level of risk with realised volatility, the square root of the sum of squared daily returns. For the dynamics we fit a GARCH(1,1) with Student- t shocks to the equity composite,

$$r_t = \mu + \varepsilon_t, \quad \varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (1)$$

where α is how strongly today’s shock feeds tomorrow’s variance and $\alpha + \beta$ is how long shocks persist. The GJR extension adds a term that lets negative shocks raise variance more than positive ones, the leverage effect. For the tail we use peaks-over-threshold: we keep daily losses above the 95th percentile and fit a Generalised Pareto Distribution, whose shape parameter ξ measures tail heaviness. We expect fat tails ($\xi > 0$) and a leverage effect ($\gamma > 0$) in both periods; the question is whether they got stronger after COVID.

3.4. Block B – dependence

For co-movement we compute, each month, the average pairwise correlation $\bar{\rho}$ across all 14 series on the trailing 60-day window. For factor concentration we run PCA on the same window and summarise it with the effective number of factors, an entropy measure

$$K_{\text{eff}} = \exp\left(-\sum_k \tilde{\delta}_k \ln \tilde{\delta}_k\right), \quad \tilde{\delta}_k = \delta_k / \sum_j \delta_j, \quad (2)$$

where the δ_k are eigenvalues; $K_{\text{eff}} = 14$ means perfectly spread risk, $K_{\text{eff}} = 1$ a single driver. Then, because an average correlation can hide a change in *how* correlation moves, we fit the DCC model of Engle (2002). After filtering each equity series through its own GARCH, DCC lets the correlation matrix evolve as

$$Q_t = (1 - a - b) \bar{Q} + a z_{t-1} z'_{t-1} + b Q_{t-1}, \quad (3)$$

where a is the speed at which correlations react to a shared shock and b is how persistent the correlation level is. EWMA (RiskMetrics, $\lambda = 0.94$) is the same recursion with a, b fixed, so we show it as the textbook special case. We expect a to rise if correlations now react faster to stress.

3.5. Block C – dynamics

Finally we study volatility and correlation together. Our main regression asks whether the sensitivity of correlation to volatility changed after COVID:

$$\bar{\rho}_t = \alpha + \beta_1 \sigma_{t-1}^{\text{eq}} + \beta_2 (\sigma_{t-1}^{\text{eq}} \times D_t^{\text{post}}) + \gamma D_t^{\text{post}} + \varepsilon_t, \quad (4)$$

where β_2 is the change in slope. Because $\bar{\rho}$ is persistent, our *primary* version adds $\bar{\rho}_{t-1}$ on the right (an ARX model), and all versions use Newey–West (Newey and West, 1987) standard errors. A placebo version replaces the COVID dummy with a generic “high-volatility” dummy, to check we are not just picking up the mechanical correlation–volatility link. We then estimate a small VAR on (volatility, correlation) for Granger causality and impulse responses, and a two-state Markov-switching model on volatility to see whether high-risk months cluster after 2020.

4. Empirical Results

The diagnostics come out as expected and set the stage. Jarque–Bera rejects normality for all 14 series, and the Ljung–Box test finds volatility clustering in 13 of them, so Student- t GARCH and extreme-value tools are the right choices. On the monthly series, ADF and KPSS agree that realised volatility is stationary but disagree on $\bar{\rho}$ (ADF $p = 0.13$, KPSS rejects): $\bar{\rho}$ is borderline and very persistent. That is exactly the case Phillips (1986) warns about, so in Block C the ARX model is our main specification and the level regression is only support.

Figure 1 shows the four monthly measures over the whole sample. The COVID line marks a spike, but nothing in the picture looks like a permanent shift in level. The interesting changes, if any, are in the dynamics, and that is where we now look.

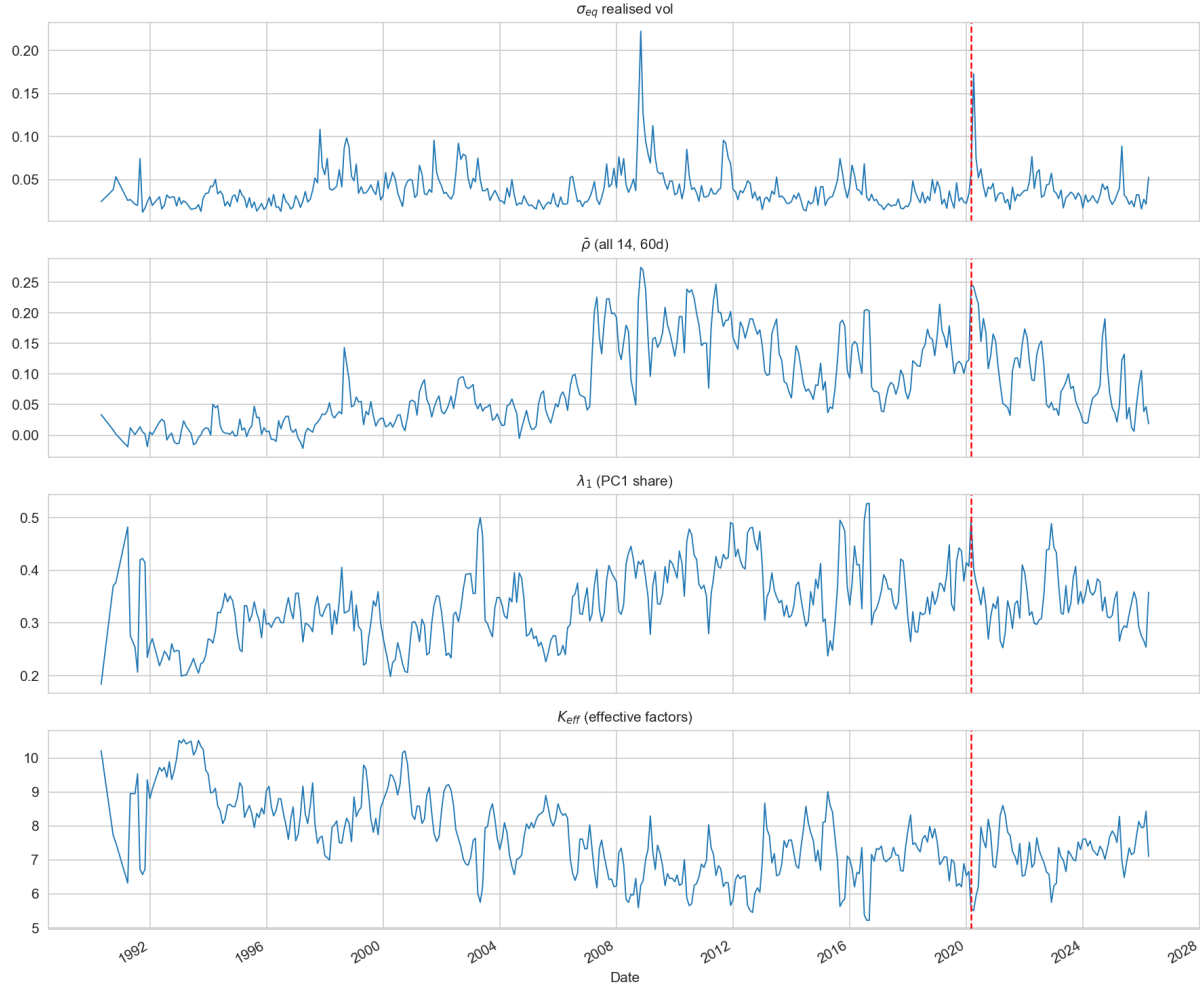


Figure 1: The four monthly risk measures, January 1990 to April 2026. From top: realised equity volatility σ^{eq} , average pairwise correlation $\bar{\rho}$, the share of variance in the first principal component λ_1 , and the effective number of factors K_{eff} . The red dashed line is March 2020.

4.1. Block A – how big and how heavy are the shocks?

The level did not move. Table 1 compares the monthly measures before and after COVID, with block-bootstrap p -values. Average realised volatility is essentially identical (0.038 vs 0.038, $p = 0.97$), and its spread is unchanged too. The violent level shock of March 2020 was real but temporary; on a monthly average it leaves no footprint. If the level is flat, the natural next question is whether the *behaviour* of volatility changed, which is what GARCH measures.

Table 1: Monthly risk measures before and after COVID, with block-bootstrap p -values.

Series	Pre mean	Recent-pre	Post mean	Pre SD	Post SD	$p(\text{mean, full})$
σ^{eq} (realised vol.)	0.038	0.032	0.038	0.022	0.021	0.966
$\bar{\rho}$ (avg. correlation)	0.081	0.118	0.090	0.069	0.056	0.639
λ_1 (PC1 share)	0.335	0.368	0.340	0.070	0.048	0.762
K_{eff} (effective factors)	7.692	7.041	7.250	1.158	0.614	0.146

Notes: Recent-pre is 2015–2020. p -values are two-sided stationary block bootstraps (blocks of 6 months) for the pre/post difference in means. A naive i.i.d. t -test would read close to zero for K_{eff} ; the bootstrap shows that is an artefact of overlapping windows. Pre-COVID: 347 months. Post-COVID: 73 months.

The dynamics did change. Table 2 reports the GARCH fits. A likelihood-ratio test rejects

equal parameters across the two periods ($LR = 15.9$, $p = 0.003$). Two things move in a sensible direction. The immediate reaction to a shock, α , is a little larger after COVID (0.107 to 0.115), and total persistence $\alpha + \beta$ falls from 0.993 to 0.957. In plain terms, shocks hit slightly harder and die out faster. We are careful here: the pre-COVID $\alpha + \beta$ sits almost at one (a near-unit-root case), so any precise “half-life” number from it would be unstable, and we are comparing 30 years against 6. We therefore read the direction, not a precise figure. The GJR results (Table 2, notes) confirm a leverage effect in both periods, slightly stronger after COVID (γ : 0.136 to 0.153), and a likelihood-ratio test that the GJR parameters changed is significant at the 5% level ($p = 0.042$). Bad news has always raised volatility more than good news; after COVID it does so a touch more.

Table 2: GARCH(1,1)- t on the equity composite (daily returns scaled by 100).

Period	$\hat{\omega}$	$\hat{\alpha}$	$\hat{\beta}$	$\hat{\alpha} + \hat{\beta}$	$\hat{\nu}$	Log-lik
Full	0.012	0.112	0.878	0.990	6.01	−10,777
Pre-COVID	0.010	0.107	0.886	0.993	6.16	−8,915
Post-COVID	0.031	0.115	0.842	0.957	5.55	−1,855

Notes: LR test of equal parameters pre/post: $LR = 15.9$, $p = 0.003$ (χ_4^2). GJR leverage term: $\hat{\gamma}_{\text{pre}} = 0.136$, $\hat{\gamma}_{\text{post}} = 0.153$ (both highly significant); LR test that the GJR parameters differ pre/post: $p = 0.042$.

The tail is probably heavier, but we cannot prove it. Shocks also have tails, so we look there directly. Table 3 shows the GPD fit on equity-composite losses. The shape parameter more than doubles (0.151 to 0.353) and the expected loss beyond the 1% worst day (CVaR) rises from 3.81% to 4.29%. That is the right sign for “heavier tails”. But honesty matters more than a headline: with only 82 post-COVID exceedances, the shape estimate is imprecise, and a likelihood-ratio test at the proper 95th-percentile threshold is *not* significant ($p = 0.39$). It only becomes significant if we lower the threshold to the 90th percentile ($p = 0.015$), which buys observations at the cost of mixing in non-tail data. So our reading is: tail risk probably increased, but six post-COVID years are not enough to confirm it at the 1% tail. The Hill plot and Q–Q plots (Appendix D) tell the same cautious story.

Table 3: Extreme-value (GPD) fit on equity-composite daily losses, 95th-percentile threshold.

Period	# losses	# exceed.	threshold u	$\hat{\xi}$	VaR _{99%}	CVaR _{99%}
Pre-COVID	7,973	399	1.45%	0.151	2.75%	3.81%
Post-COVID	1,626	82	1.31%	0.353	2.64%	4.29%

Notes: VaR is slightly lower post-COVID because the post threshold is lower; the heavier tail shows up in the higher CVaR. LR test at a common 95th-percentile threshold: $p = 0.39$ (68 post exceedances, below the rule-of-thumb of 100). At the 90th percentile: $p = 0.015$.

So the marginals changed in part: the level is flat, the dynamics shifted, the tail is suggestive but unproven. The obvious next question is whether assets now move together more, which would matter even more for a diversified portfolio.

4.2. Block B – does diversification still work?

Average correlation is flat; concentration tightened only a little. Back in Table 1, average correlation $\bar{\rho}$ is statistically the same before and after ($p = 0.64$). The effective number of factors K_{eff} falls from 7.69 to 7.25, which looks like a loss of diversification, but under the honest block bootstrap that drop is *not* significant ($p = 0.15$). What *is* significant is that its

variability shrank ($p = 0.02$): after COVID the market sits more consistently at a slightly more concentrated structure, rather than at a clearly lower number of factors. The static PCA agrees with this modest reading: seven principal components were needed to explain 80% of variance before COVID, six after (Figure 2). This is a small narrowing, not a collapse, and we report it as such. It also raises a sharper question: if the *average* correlation did not move, could the *way* correlations move have changed?

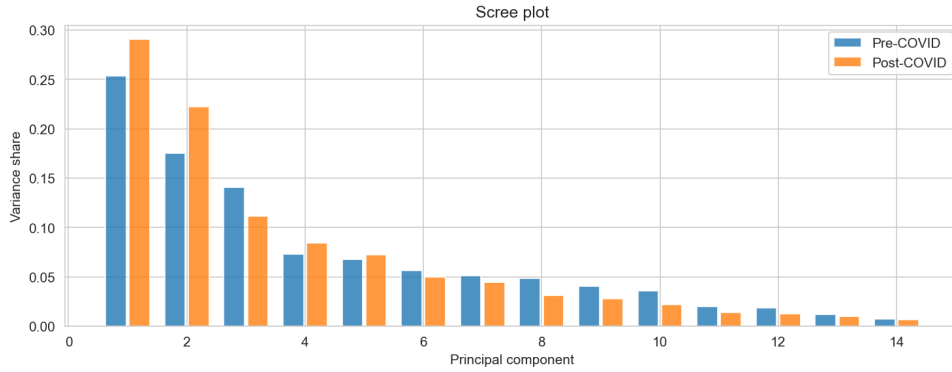


Figure 2: Share of variance explained by each principal component, pre- vs post-COVID. Seven components reach 80% before COVID, six after, a modest tightening.

Correlations now react faster. This is the clearest result in the paper, and it comes from DCC. Table 4 shows the reaction parameter a rising from 0.010 to 0.016 and the persistence parameter b falling from 0.988 to 0.958, with a likelihood-ratio test that strongly rejects equal dynamics ($\text{LR} = 57$, $p < 0.001$). The reading is intuitive: after COVID, a shared shock pushes correlations up faster, and the old level matters a little less. Figure 3 plots the dynamic correlation against the simple rolling average; both jump in March 2020, but the dynamic measure reacts and then comes back more sharply. This is the change a single average could never show, and it is the reason the average looked flat.

Table 4: DCC parameters on the five equity indices (two-step estimation).

Period	$\hat{a}$ (reaction)	$\hat{b}$ (persistence)	$\hat{a} + \hat{b}$
Full	0.010	0.987	0.997
Pre-COVID	0.010	0.988	0.998
Post-COVID	0.016	0.958	0.975

Notes: LR test of equal (a, b) pre/post: $\text{LR} = 57$, $p < 0.001$ (χ^2_2). EWMA ($\lambda = 0.94$) is the same recursion with $a = 0.06$, $b = 0.94$ fixed; we use it as a cross-check (Figure 3).

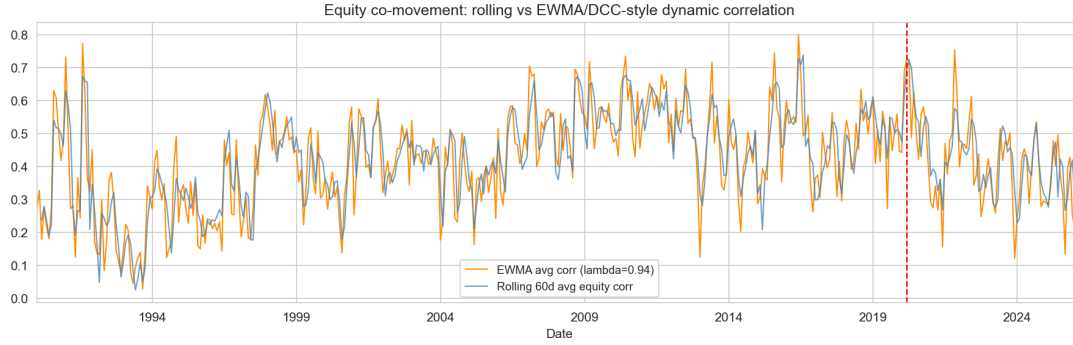


Figure 3: Average equity correlation: the 60-day rolling average (blue) and the DCC/EWMA dynamic measure (orange). Both spike in March 2020; the dynamic measure reacts and reverts more sharply.

So dependence did change, but in the *speed* of correlation, not in its average level. That sets up the last block. If correlations react faster, does the static link between volatility and correlation, and the feedback between them, look any different?

4.3. Block C – how do shocks travel and persist?

The static volatility–correlation slope did not change. Table 5 reports the main regression. In our preferred ARX specification, the COVID interaction β_2 is tiny and far from significant ($p = 0.79$), and a joint test on the COVID terms gives $p = 0.76$. We are careful with words here: this is “we could not detect a change”, not “the change is zero”. With 73 post-COVID months the test has little power, and Block B already showed the real change is in the *speed* of correlation, which a single slope cannot capture. The result holds across alternative breakpoints, a 40-day window, and a first-difference version. The placebo (Model G) is the useful check: replacing the COVID dummy with a generic high-volatility dummy gives a large, significant *negative* interaction ($\beta_2 = -2.11$, $p < 0.001$). That is the familiar concave shape from Longin and Solnik (2001): when volatility is already high, correlations sit near their ceiling and extra volatility adds little. It confirms that the volatility–correlation link is mechanical, and that COVID did not bend it.

Table 5: Correlation–volatility regression (dependent variable $\bar{\rho}_t$), Newey–West errors.

	Model A	Model F	ARX-F (main)	Placebo G
Constant	0.034***	0.034***	0.006***	−0.016
$\sigma_{t-1}^{\text{eq}} (\beta_1)$	1.301***	1.265***	0.112*	3.051***
$\sigma_{t-1}^{\text{eq}} \times D^{\text{post}} (\beta_2)$	—	0.216	0.031	—
$D^{\text{post}} (\gamma)$	—	0.001	−0.003	—
$\sigma_{t-1}^{\text{eq}} \times D^{\text{stress}}$	—	—	—	−2.108***
$\bar{\rho}_{t-1} (\phi)$	—	—	0.886***	—
Adj. R^2	0.171	0.170	0.812	0.202
N	420	420	420	420

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. ARX-F is the main specification (it adds $\bar{\rho}_{t-1}$ and guards against spurious regression). Joint Wald test on the COVID terms in ARX-F: $p = 0.76$. The full A–H battery is in Appendix B.

The feedback loop is intact. A VAR with four lags (chosen by Hannan–Quinn) shows two-way Granger causality: volatility helps predict correlation ($F = 4.89$, $p < 0.001$) and correlation helps predict volatility ($F = 4.14$, $p = 0.003$). The impulse response (Figure 4) says a volatility shock raises correlation for a few months and fades by about month six, and a variance decomposition

attributes 14% of correlation's forecast error to volatility at one month, peaking near 22% at three months. The pre-COVID Granger link is clearly significant ($p = 0.006$); the post-COVID one is only borderline ($p = 0.064$), but with 73 months that is what limited power looks like, so we do not lean on a pre/post difference here. The transmission channel between volatility and correlation looks the same after COVID as before.

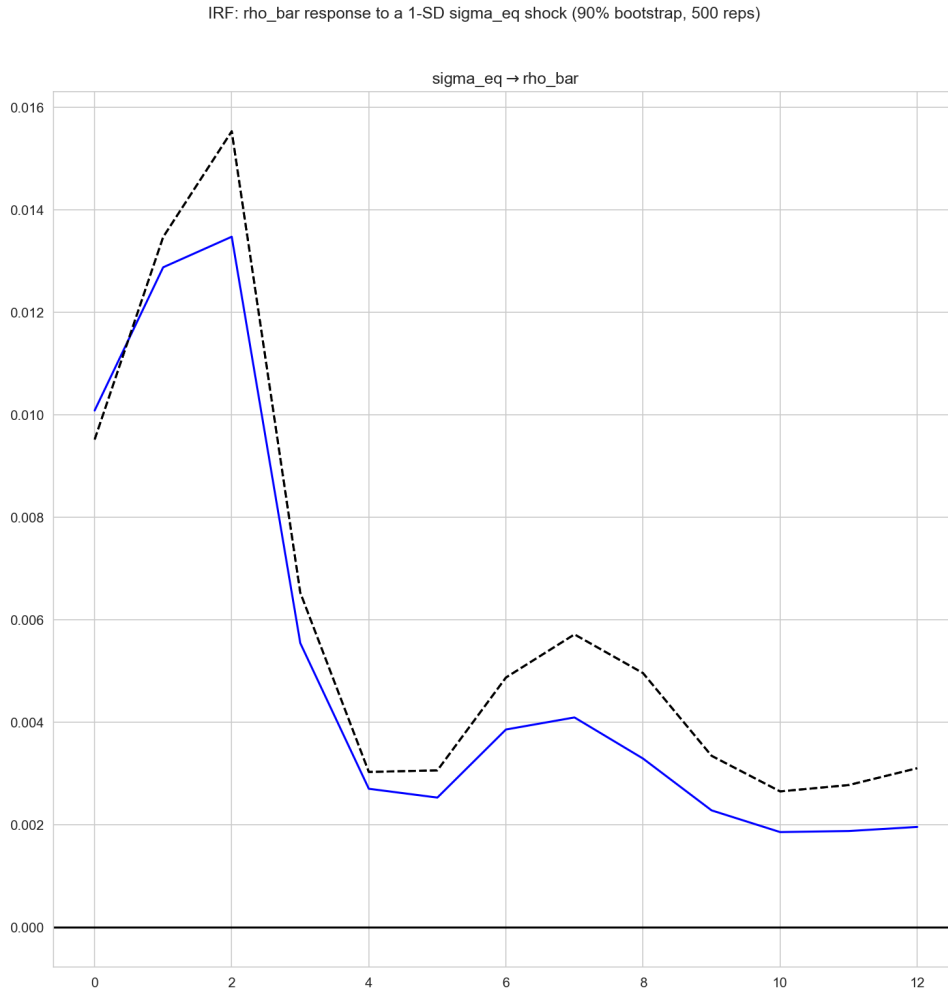


Figure 4: Response of average correlation to a one-standard-deviation volatility shock (VAR, Cholesky ordering with volatility first). The effect peaks early and fades within about six months. Shaded band: 90% bootstrap interval.

Regimes: a weak and benchmark-dependent signal. The two-state Markov-switching model on monthly volatility finds a calm regime (mean 0.029) and a stressed one (mean 0.065). Whether post-COVID months sit more often in the stressed regime depends entirely on the benchmark. Measured against the full 30-year past, which includes 2008, the post-COVID period spends *less* time in stress (19.6% vs 26.3%). Measured against the recent 2015–2020 calm it spends slightly *more* (19.6% vs 18.1%), a gap of about 1.4 percentage points. We will not dress up a 1.4-point, benchmark-dependent difference as a structural change. On its own, the regime evidence is too weak to support a clear conclusion (Figure 9, Appendix E).

4.4. Putting it together

Table 6 collects the verdict, one row per feature, with the honest confidence level. Two features clearly moved (volatility dynamics, correlation reaction speed), two clearly did not (volatility

level, the static slope), and three are weak or unproven (tail risk, factor concentration, regimes). The pattern is consistent: what changed is *how fast* risk behaves, not its average level.

Table 6: Did it change? A summary across the features of the risk structure.

Feature	Changed?	Main evidence	Confidence
Volatility level	No	realised vol mean flat ($p = 0.97$)	High
Volatility dynamics	Yes	GARCH LR break; $\alpha \uparrow$, $\alpha + \beta \downarrow$	Medium
Tail risk	Probably	ξ , CVaR up; few exceedances	Low
Factor concentration	Partly (variance)	K_{eff} mean flat ($p = 0.15$); variance down	Low
Correlation dynamics	Yes	DCC $a \uparrow$, $b \downarrow$; LR break	Medium-high
Vol.-corr. static slope	Not detected	β_2 insignificant everywhere	Limited power
Regime frequency	Unclear	benchmark-dependent sign	Low

5. Discussion

What does this mean for someone actually managing risk? The short version is that the parts of a risk model worth updating are the ones describing *how fast* risk moves, not the ones describing its average level.

Start with volatility. Persistence fell, so a shock fades faster than a pre-2020 model assumes. A desk still running old GARCH parameters keeps its risk estimates high for too long after a shock and over-hedges well into the recovery. Correlation is the subtler case. Its average did not move, yet it now reacts faster, so a model that updates correlation slowly arrives late to a stress episode. A conditional, DCC-style correlation reacts in time; a long rolling window does not. One warning follows from the same results. Because the static slope did not change, a given volatility spike does not produce a *bigger* correlation jump than before, only a *faster* one, and that difference is easy to miss in a stress test.

Diversification is the mild part of the story. The number of independent risk factors did not fall in a way we can defend statistically; it only grew more stable. A manager need not tear up the diversification plan, but should expect a little less day-to-day movement in how concentrated the market feels. The tail is the open question. The signal points to heavier losses, but it is not strong enough yet to act on with confidence, so it belongs on a watch list rather than in a rewritten capital model.

6. Conclusion

Did the structure of risk change after COVID-19? Our answer is *partially, and through specific channels*. Volatility shocks became a little sharper and shorter-lived, and cross-asset correlations now react faster to shared shocks; both changes are statistically clear. The average level of volatility and of correlation did not move, and the static link between them did not change. Tail risk and factor concentration point in the “riskier” direction but are too weak, on six post-COVID years, to confirm, and the regime evidence depends on the benchmark.

We are upfront about the limits. The post-COVID window is only 73 months, which caps the power of every test; we use one equity composite rather than studying each asset class separately; and by design we use no macro or policy variables, so we describe the change without explaining

its cause. The honest way to keep this question alive is to watch two numbers as the sample grows: GARCH persistence $\alpha + \beta$ and the DCC reaction parameter a . If they drift back toward their pre-2020 values, the changes we document were temporary; if they hold, the structure of risk really did move, and the time to recalibrate is now.

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A. Stationarity tests

Table 7: ADF and KPSS tests on the monthly risk measures.

Series	ADF stat	ADF p	ADF decision	KPSS p	KPSS decision
σ^{eq}	-5.66	< 0.001	stationary	≥ 0.10	stationary
$\bar{\rho}$	-2.46	0.126	unit root?	≤ 0.01	non-stationary
λ_1	-4.12	0.001	stationary	≤ 0.01	non-stationary
K_{eff}	-2.99	0.036	stationary	≤ 0.01	non-stationary

Notes: The disagreement for $\bar{\rho}$ (ADF cannot reject a unit root, KPSS rejects stationarity) is why we use the ARX and first-difference specifications in Block C. KPSS p -values are reported on a discrete grid.

B. Full regression battery

Table 8: Models A–H, dependent variable $\bar{\rho}_t$, Newey–West errors.

	A	B	C	D	E	F	G	H
σ_{t-1}^{eq}	1.301***	1.857***	1.477***	1.830***	2.753***	1.265***	3.051***	2.307***
$\times D^{\text{post}}$	—	—	—	—	—	0.216	—	0.149
D^{post}	—	—	—	—	—	0.001	—	-0.020
$\times D^{\text{stress}}$	—	—	—	—	—	—	-2.108***	—
Block B controls		✓			sel.			✓
Block C controls			✓		sel.			✓
Block D controls				✓	sel.			✓
Adj. R^2	0.171	0.287	0.253	0.261	0.358	0.170	0.202	0.409
N	420	420	420	420	420	420	420	420

*** $p < 0.01$. Block B: peer volatilities (oil, gold, EM, Hang Seng). Block C: credit spread, transatlantic yield spread, change in US 10Y. Block D: FX volatilities. “sel.” keeps the most significant control from each block. The COVID interaction β_2 is insignificant in every model that contains it; the only significant interaction is the placebo stress term in G.

Additional robustness (all consistent with no detectable slope change): alternative breakpoints 20 February 2020 ($\beta_2 = 0.19$, $p = 0.50$) and 1 April 2020 ($\beta_2 = 0.12$, $p = 0.67$); a first-differenced dependent variable ($\beta_2 = 0.01$, $p = 0.95$); and a 40-day rolling window. Using λ_1 as the dependent variable does show a significant interaction ($\beta_2 = -0.82$, $p = 0.02$), the one place a COVID slope effect appears.

C. Sub-block correlations and PC1 loadings

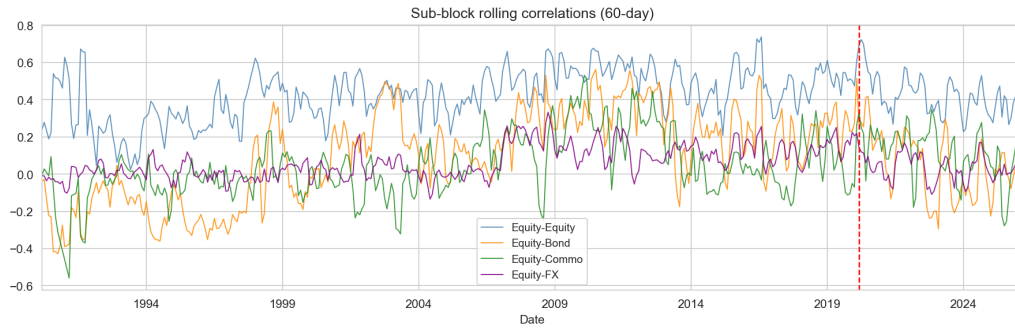


Figure 5: Rolling 60-day correlation between equities and each other asset class.

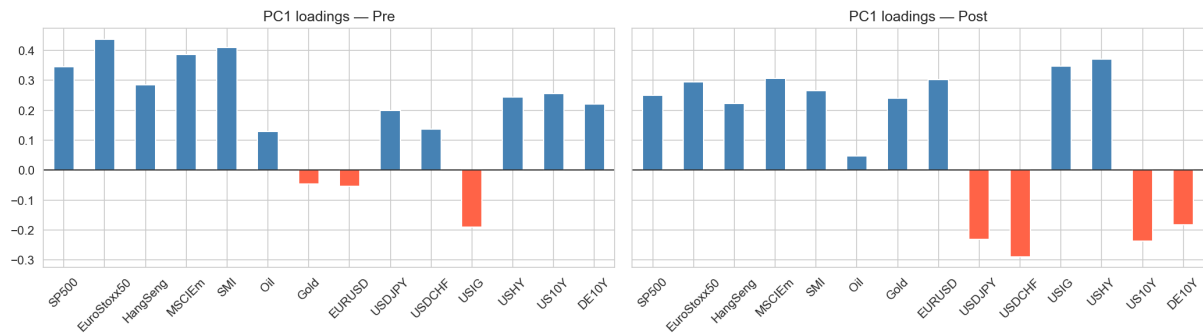


Figure 6: Loadings on the first principal component, pre- vs post-COVID (sign fixed so equities load positively).

D. Tail diagnostics

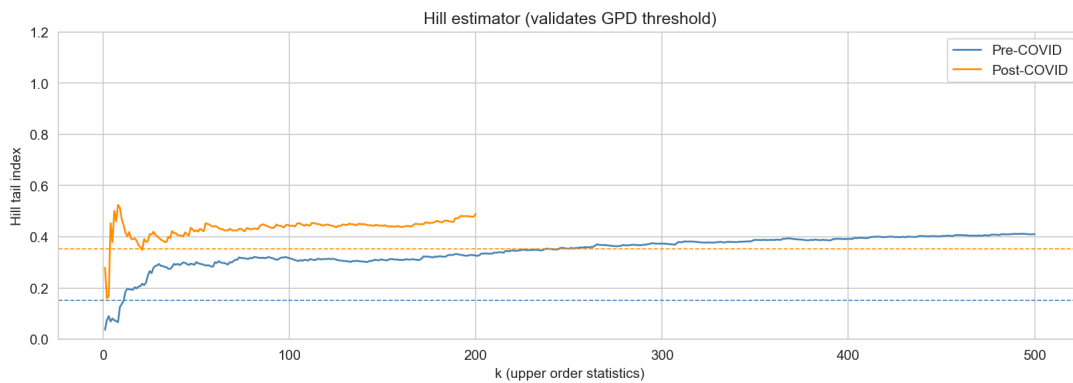


Figure 7: Hill tail-index estimator against the number of upper order statistics. The post-COVID curve sits higher, but the short sample makes the level uncertain.

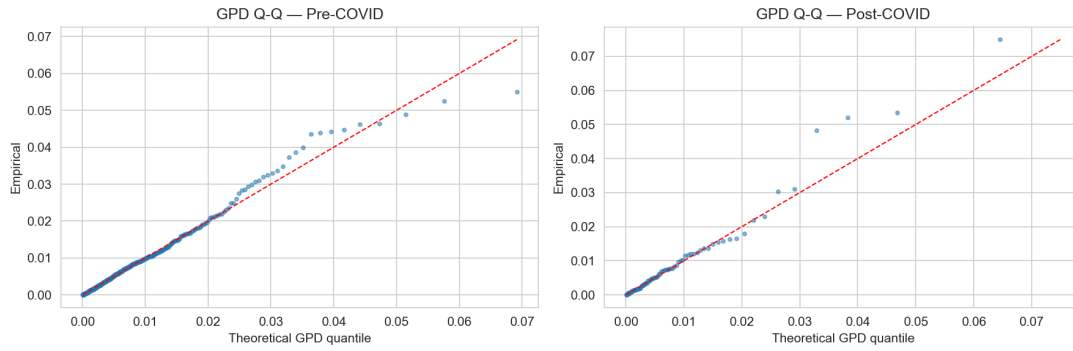


Figure 8: Q–Q plots of exceedances against the fitted GPD, pre (left) and post (right). The points track the 45-degree line, so the GPD fit is reasonable in both periods.

E. Regimes and transmission

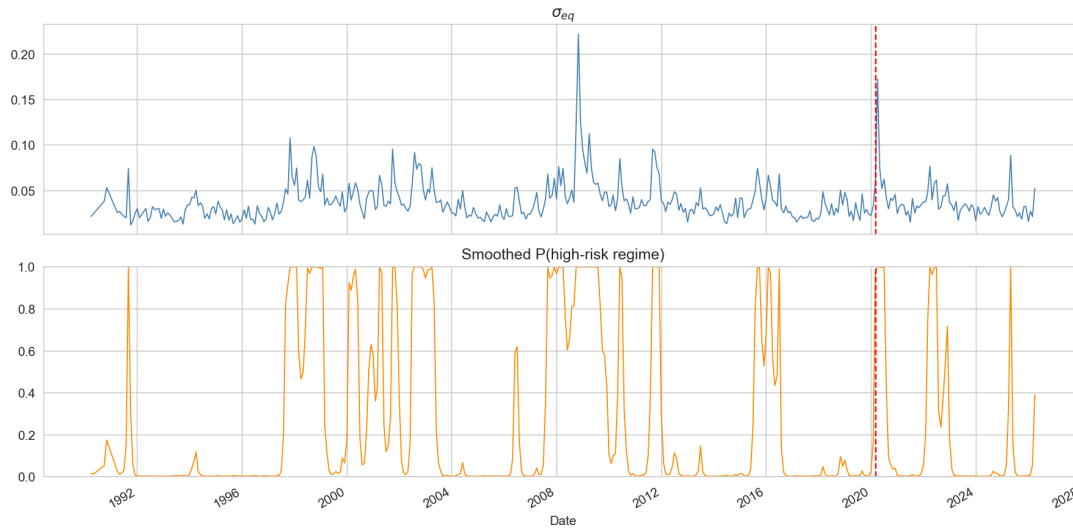


Figure 9: Top: monthly realised volatility. Bottom: smoothed probability of the high-risk regime from the two-state Markov-switching model. The red line is March 2020.

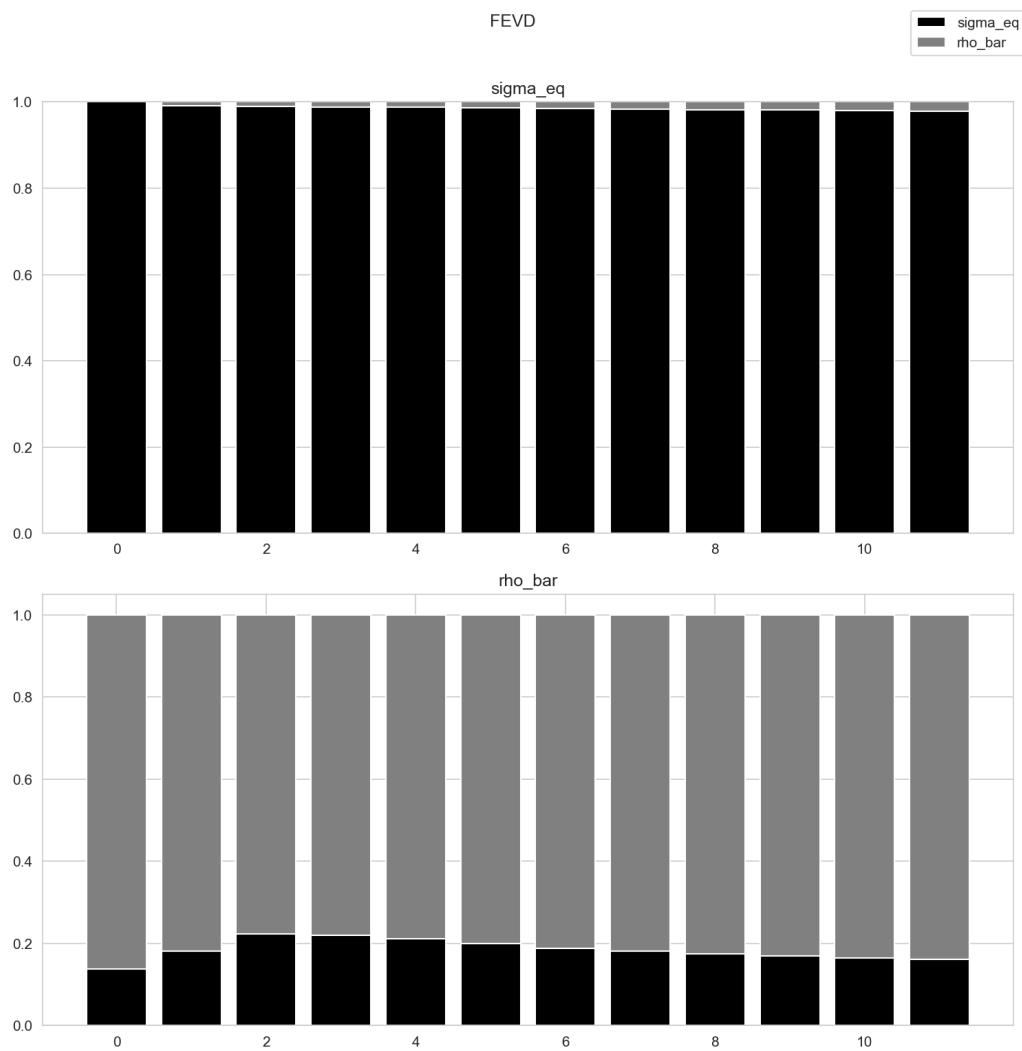


Figure 10: Forecast-error variance decomposition for the VAR on (volatility, correlation).

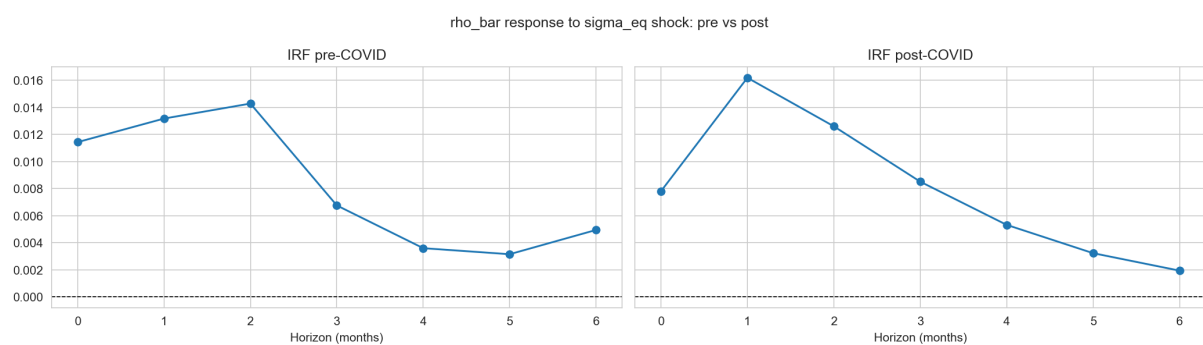


Figure 11: Correlation's response to a volatility shock, estimated separately on the pre- and post-COVID samples. The post-COVID estimate is noisy (73 months) and we do not read a pre/post difference into it.